Bluefish (*Pomatomus saltatrix*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low

Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.12	2.9	
Complexity in Reproductive Strategy	1.44	2.95	
Dispersal of Early Life Stages	1.52	2.8	
Egg and Larval Survival and Settlement Requirements	1.76	3	
Historical Variability in Temperature	1.4	2.95	
Juvenile Habitat Requirements	3.96	3	
Other Stressors	2.44	2.8	
Population Growth Rate	2.16	2.8	
Prey Specificity	1.56	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.08	2	
Spawning Cycle	1.88	3	
Stock Size/Status	2	3	
Total Sensitivity | Certainty	**Low (1) | 0.69**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	3.01	1.18	
Sea Surface Salinity	3.17	0.90	
Surface pH	3.87	0.40	
Mixed layer depth (delta rho = 0.03)	1.14	0.47	
Small zooplankton integrated biomass	1.46	0.52	
Medium zooplankton integrated biomass	1.39	0.47	
Large zooplankton integrated biomass	1.06	0.13	
Total Exposure	**High (3.11)**	**0.69**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (3.11)**	**0.69**	

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Bluefish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	High	High	None
Vulnerability	Low	Low	None
Directional Effect	Positive	Neutral	Decrease

Stocks

Bluefish is managed as a single stock or stock information was not considered for this iteration of the CVA.
Score Descriptions:

Sensitivity:

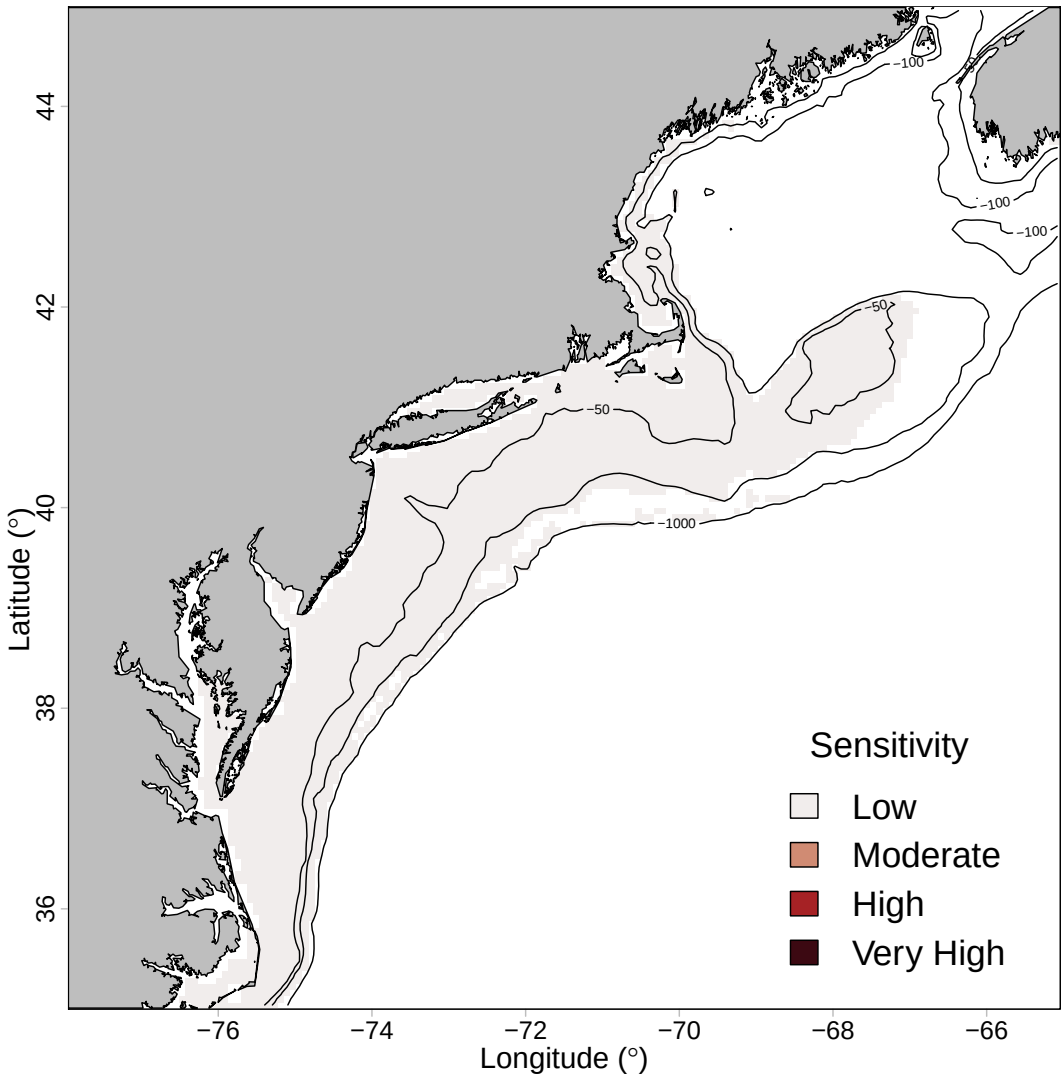


Figure 2: Bluefish Sensitivity on the Northeast Shelf

Bluefish received a **Low** sensitivity score, with a certainty of 69% (Figure 1). The only attribute that received a score greater than 2.5 was Juvenile Habitat Requirements (4) (Figure 1). Bluefish rely entirely on salt marshes and submerged aquatic vegetation as juveniles; both habitats are considered highly vulnerable to future climate change (Guyant et al. 2022).

Bluefish also received a Low sensitivity score in the 2016 Assessment. Similar to our results, Early Life History Requirements was the only attribute to score above 2.5 (Hare et al. 2016).

Exposure:

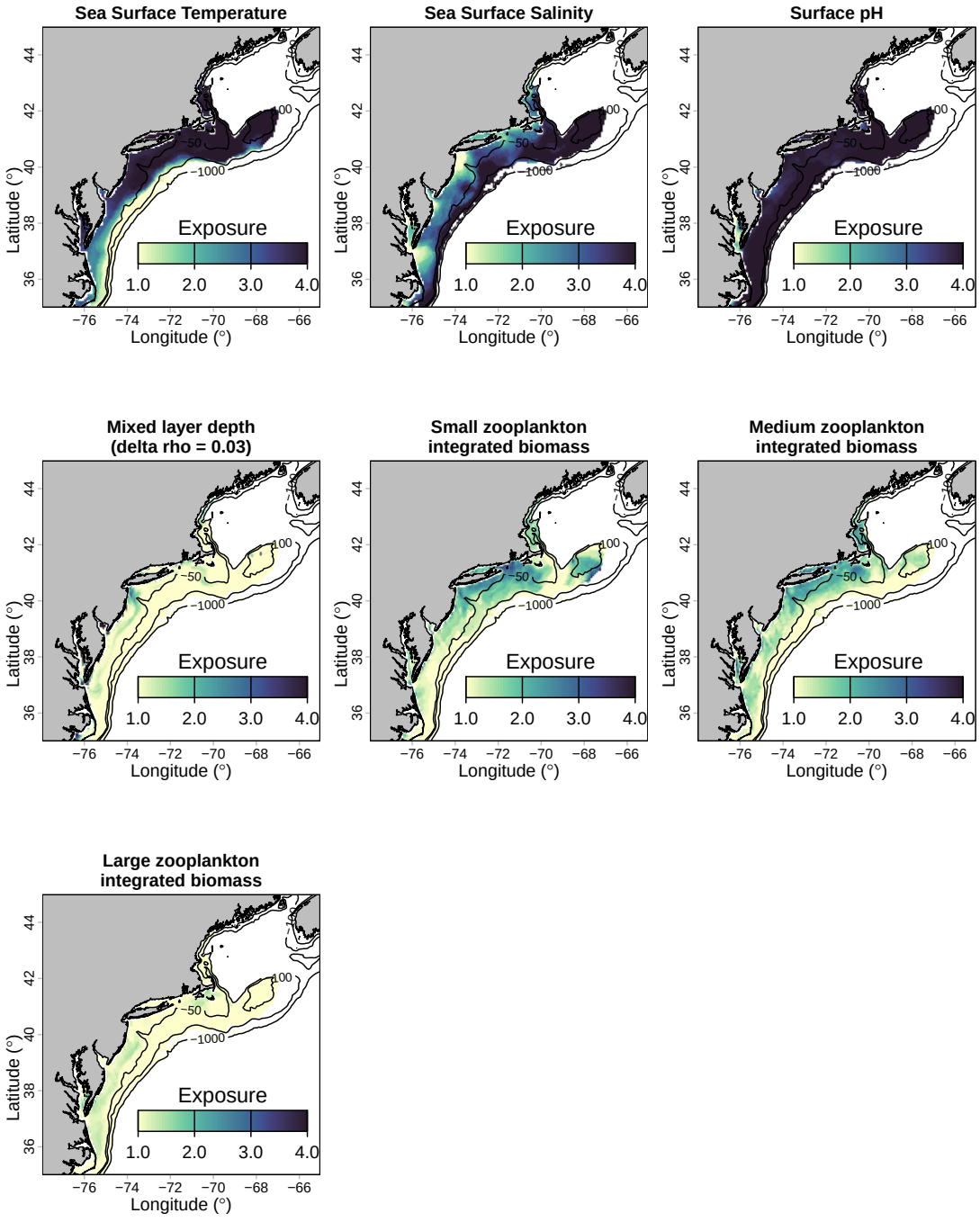


Figure 3: Maps of each Exposure Factor for Bluefish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure was **High** (3.11 +/- 0.69) (Figure 1). Bluefish exposure was Very High along Georges Bank and south of Cape Cod, and Low off the coast of Chesapeake Bay (Figure 4). These patterns were

driven by high exposure to changes in sea surface temperature, salinity, and pH (Figure 3).

In the previous Assessment, Bluefish also received a High Exposure score due to high exposure to sea surface temperature and ocean acidification (Hare et al. 2016).

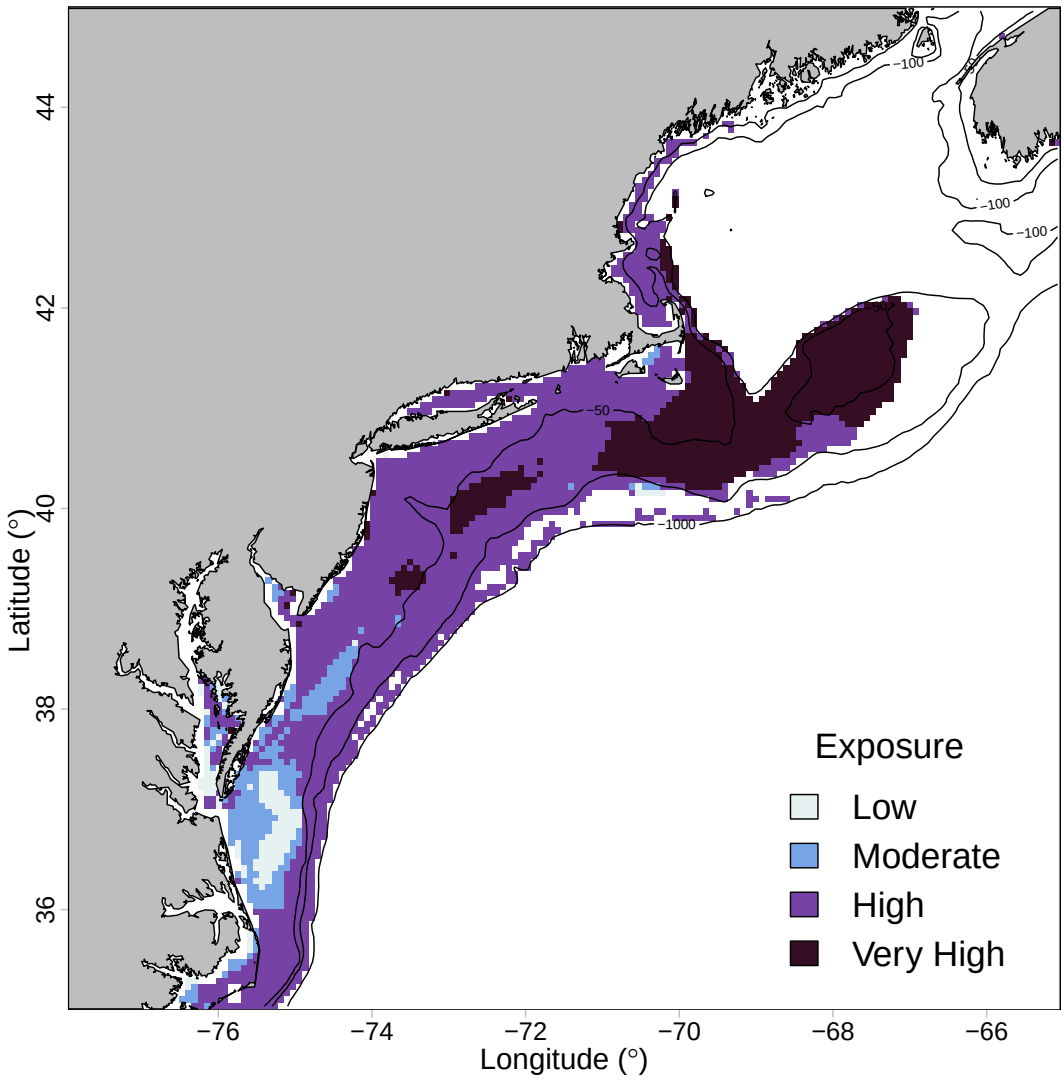


Figure 4: Total Exposure on the Northeast Shelf for Bluefish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

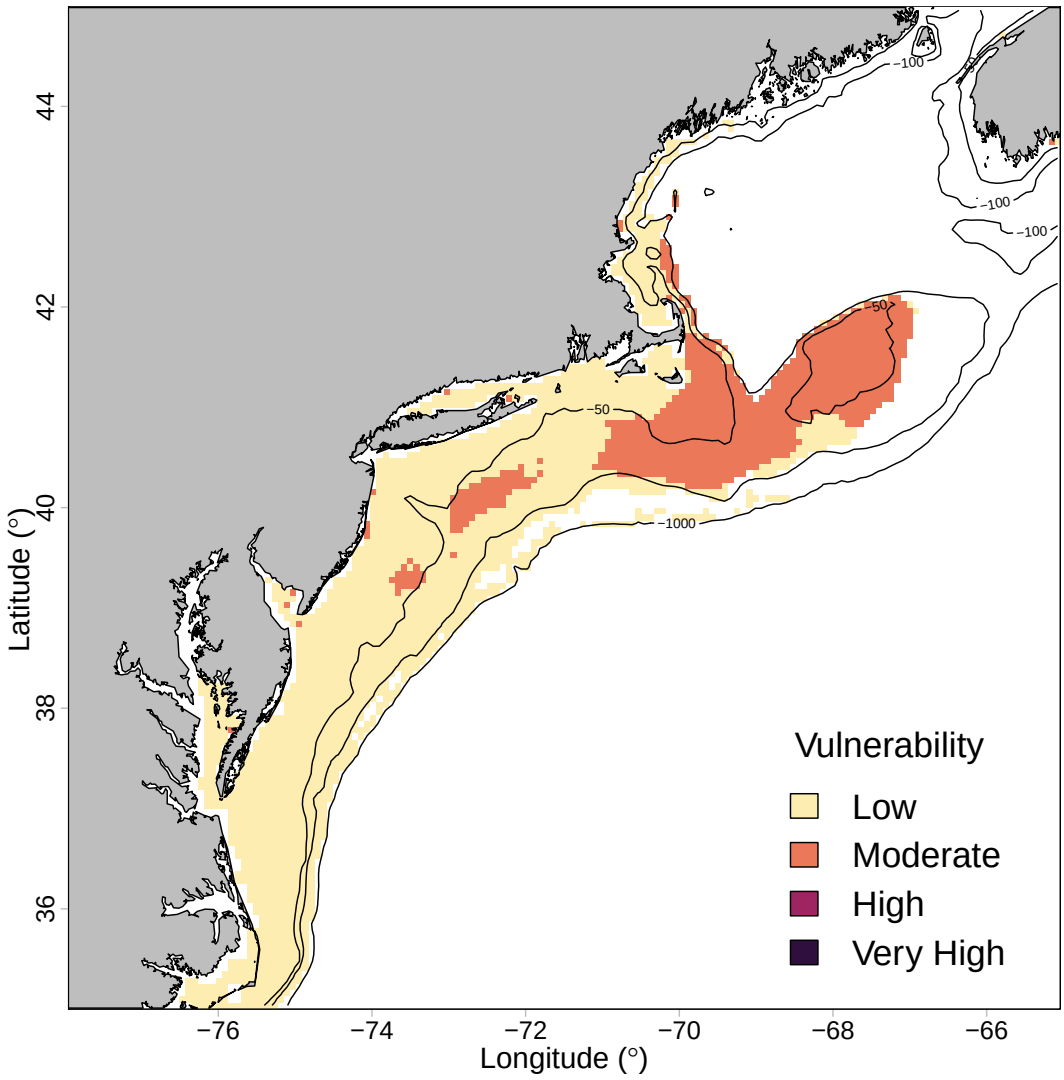


Figure 5: Vulnerability on the Northeast Shelf for Bluefish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean Vulnerability for Bluefish was **Low** (3.11 +/- 0.69) due to Low Sensitivity scores (Figure 1). Vulnerability was Moderate along Georges Bank and south of Cape Cod but Low throughout the rest of their range (Figure 5).

In the previous Assessment, Bluefish also received a Low Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

97% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The bluefish mean model confidence score was 2.6 (+/- 0.3), indicating high confidence in the model. Experts stated that the final model captures the species distributions well. There were spatial patterns in residuals, particularly during the fall migration, but this behavior varies with interannual shifts in temperature, so residuals were expected. The ensemble model AUC was 0.95 indicating that the model had excellent prediction skill.

Additional Metrics

Distribution Shifts:

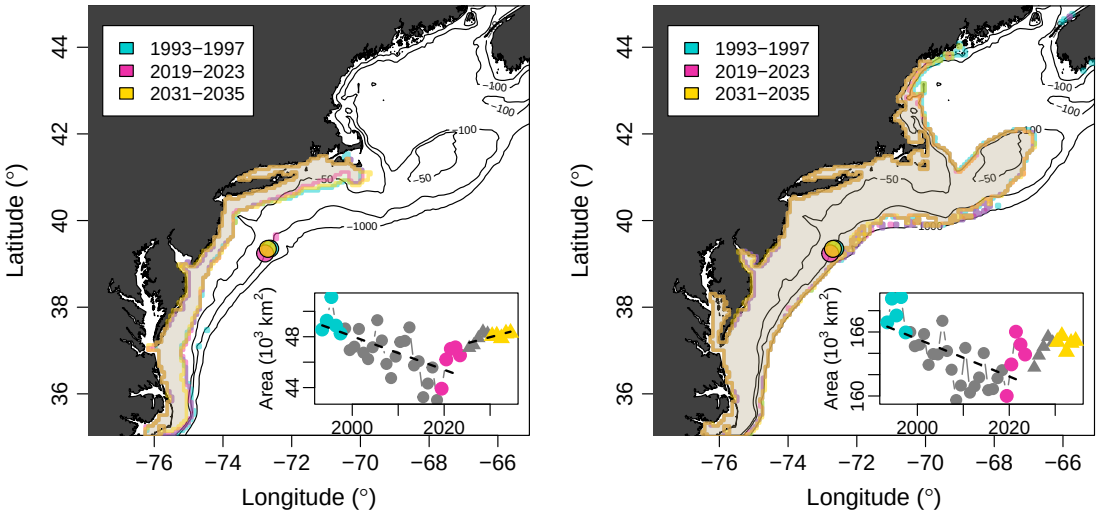


Figure 6: Distribution shifts on the Northeast Shelf for Bluefish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Bluefish distributions contracted closer to shore between 1993 and 2023, resulting in significant habitat area losses. The northern edge of bluefish's range also shifted southward (Figure 6). During this time, the weighted centroid shifted 18 km southwest.

Under near future climate conditions, bluefish core habitat area is predicted to increase to near 1993 levels (Figure 6). The weighted centroid is predicted to shift 14 km northeast, back towards the 1993 location.

In the previous Assessment, bluefish received a High Distributional Vulnerability Rank due to high adult mobility, large seasonal migrations, and high larval dispersal (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	49,512	167,963	6.80	7.86
2019-2023	17.6 km SW	46,260 (-3,252)	163,618 (-4,344)	6.42 (-0.37)	7.94 (+0.08)
2031-2035	13.5 km NE	48,114 (+1,854)	164,983 (+1,364)	6.42 (0)	7.94 (0)

Directional Effect:

Bluefish are expected to be **neutrally** impacted by climate change. Expert certainty was 95%. Distributions are shifting north and east and warming temperatures may expand the spawning season (Tyrell et al. 2022). However, there is a concern that seasonal mismatches may occur if juveniles arrive in estuaries before their prey, which may impact recruitment (Taylor and Able 2006). In the original Assessment, bluefish were predicted to be positively impacted by climate change due to expanding distributions and spawning seasons (Hare et al. 2016).

Climate Effects on Abundance and Distribution

There has been a documented shift in bluefish distribution, spawning, habitat use in the Mediterranean (Sabatés et al. 2012) (Villegas-Hernández et al. 2015). Bluefish distributions in the Northwest Atlantic seem to be shifting north and east (Tyrell et al. 2022). Increasing fall sea surface temperatures may extend spawning season further into the calendar year (Tyrell et al. 2022). If warming alters spawning phenology, there are concerns that juveniles may arrive in estuaries before their prey. Seasonal mismatches have been shown to impact growth, body condition, and reproductive success (Taylor and Able 2006). Changing distributions of prey species as a result of climate change may alter interactions between bluefish and their prey (Whitfield 2026).

Life History Synopsis

Bluefish inhabit temperate and tropical surface waters in the North Atlantic from Maine to the Atlantic coast of Florida (Shepherd and Packer 2006). Bluefish in the North Atlantic are a single, migratory stock with two discrete spawning periods. The first occurs during March-May near the edge of the continental shelf of the South-Atlantic Bight. The second occurs between June and August in the Mid-Atlantic Bight (Shepherd and Packer 2006). Eggs hatch after 46-48 hours at 18 - 22.2°C to pelagic larvae (Deuel et al. 1966). Larvae spawned in the South Atlantic (spring-spawned cohort) are subject to advection north via the Gulf Stream and inshore via warm core rings. Eggs in the southern part of the Mid-Atlantic Bight may be advected south and offshore (Shepherd and Packer 2006). Transport of larvae and/or juveniles across the shelf- slope region may be aided by wind-driven surface flow (Hare and Cowen 1996). Sherman et al. (1984) note that wide distribution of bluefish larvae over the shelf reduces their vulnerability to the predator field and environmental perturbation. Larvae prey on copepods (Shepherd and Packer 2006). Peak abundance of larval bluefish in Southern New England and the Mid-Atlantic Bight coincided with zooplankton production (Sherman et al. 1984). Changes in prey availability and distribution can impact

bluefish condition and reproductive success (Taylor and Able 2006). Juvenile bluefish inhabit estuaries, bays, and coastal oceans with sand, oyster beds, and submerged aquatic vegetation habitats (McBride and Conover 1991). Juveniles enter estuaries when temperatures are greater than 13-15°C and leave when temperatures fall below 10-12°C (Smith et al. 1994). Juvenile migration into estuarine habitats appears tied to a shift in eating other fish. In estuaries, adults and juveniles eat whatever taxa are locally abundant (fish, crustaceans, shrimp, and polychaetes). In ocean habitats, young-of-the-year bluefish switch to piscivory predation with increasing size, similar to estuarine habitats (Marks et al. 1996). Bluefish are opportunistic foragers, and the composition of adult bluefish diet depends largely on the available prey, but they consume similar prey items as juvenile bluefish (Shepherd and Packer 2006). Bluefish have long been recognized as a notable predator of forage fish (Tyrell et al. 2022). Adult bluefish conduct seasonal migrations following warmer waters, overwintering in the south and moving north in the spring and summer (Shepherd and Packer 2006). Migrations may be cued by photoperiod as well, and have been observed to shift northward in warm years (Nye et al. 2009). Shortfin mako sharks are identified as a major predator of bluefish and may consume more individuals than removed by the fishery (Wood et al. 2009). Most bluefish are mature by age 2, with approximately half maturing by age 1 (Northeast Fisheries Science Center (NEFSC) 2015). The maximum age for bluefish in the North Atlantic is 12 years (Salerno et al. 2001). In the Northwest Atlantic, bluefish are managed by the Mid-Atlantic Fishery Management Council and Atlantic States Marine Fisheries Commission. The last stock assessment occurred in 2025, where bluefish were found to not be overfished nor is overfishing occurring (Northeast Fisheries Science Center (NEFSC) 2025).

Supplemental Scores

Exposure Timeseries

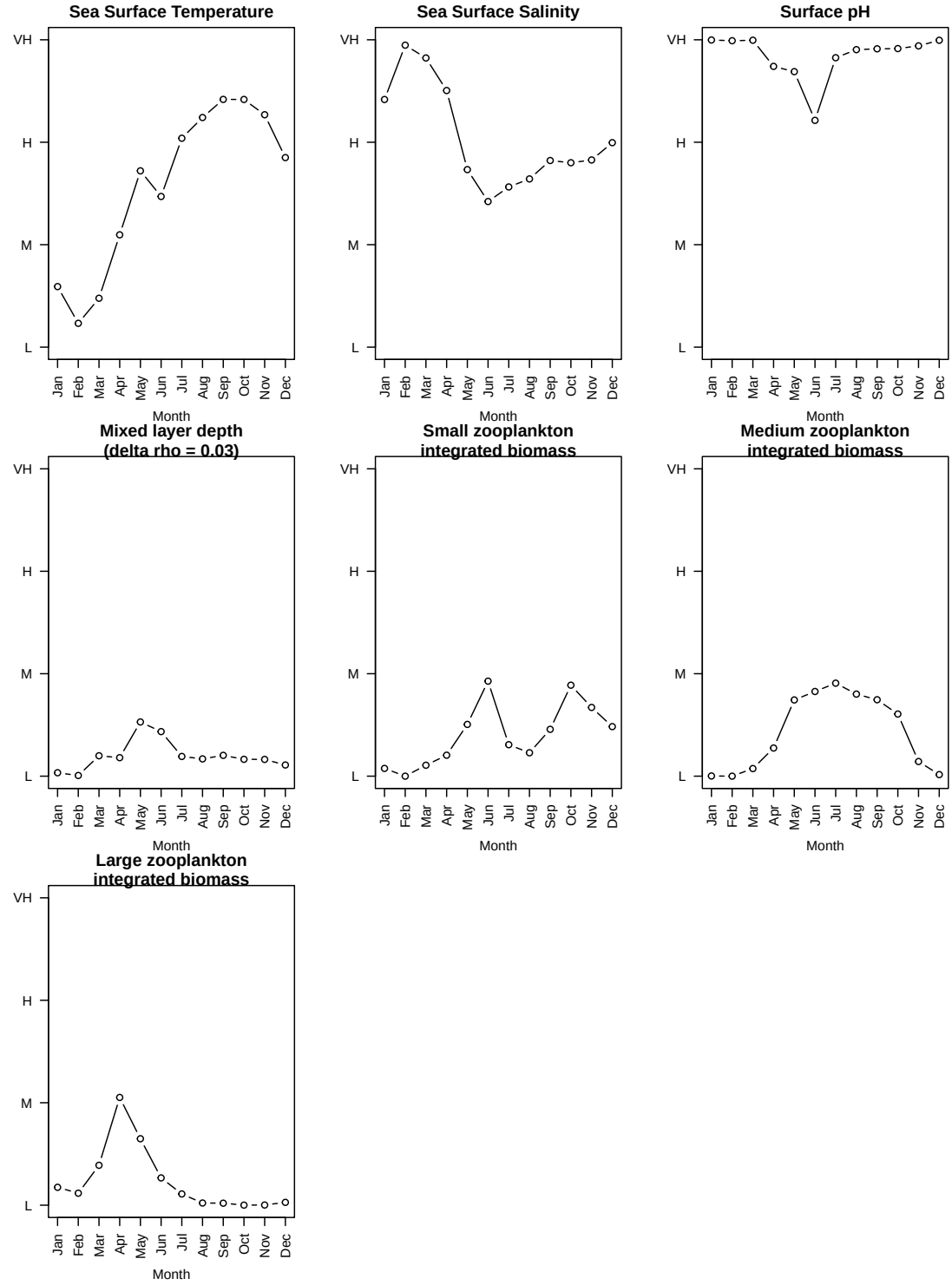


Figure 7: Monthly Timeseries of each Exposure Factor for Bluefish. Exposure was calculated between 2014-2023 and 2025-2035

There was little month-to-month variability in exposure for bluefish (Figure 8). Exposure was mostly High month to month, but declines in June and December due to declines in salinity exposure (Figure 7).

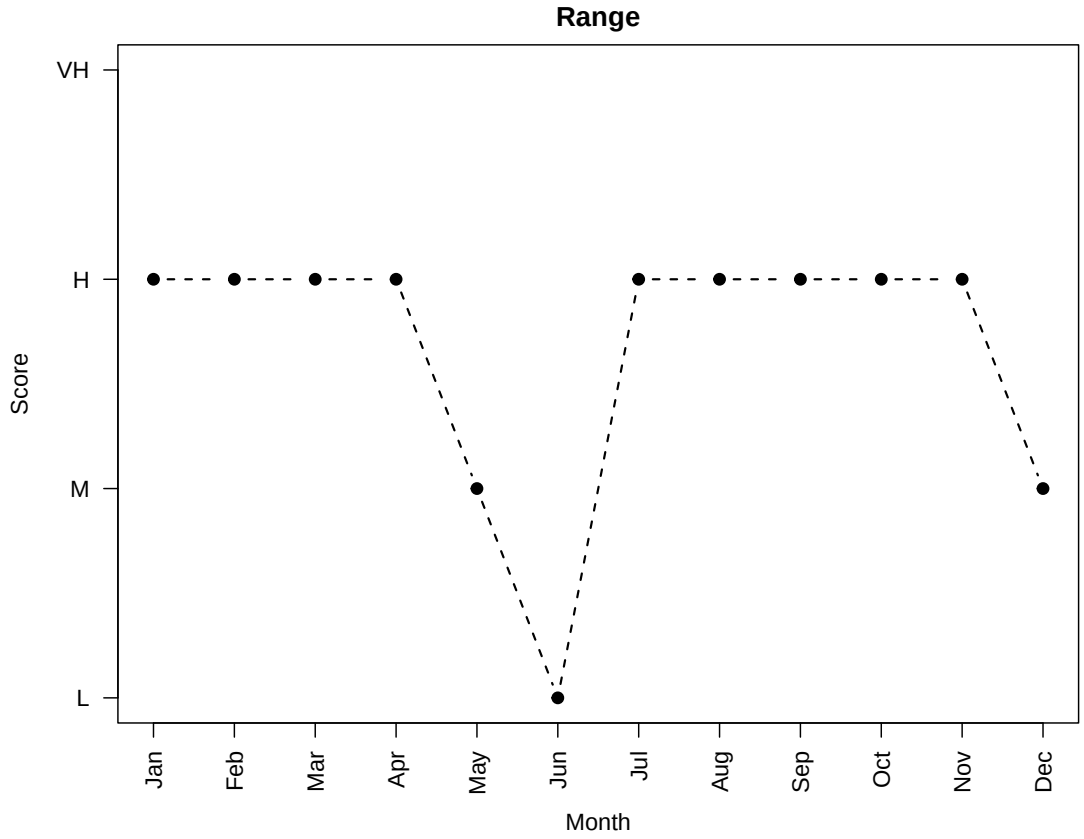


Figure 8: Timeseries of Total Exposure for Bluefish

Exposure due to Significant Factors

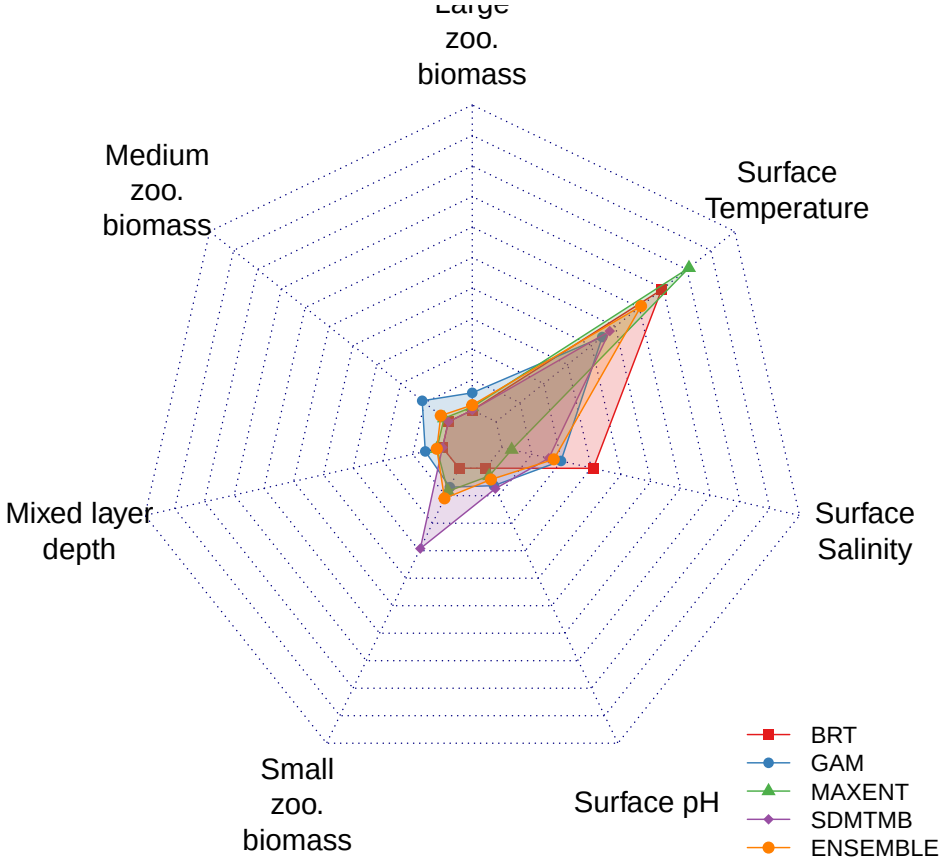


Figure 9: Dynamic Factors driving Bluefish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperature, surface salinity, and small zooplankton biomass were the major drivers of bluefish distribution (Figure 9). While adult bluefish are piscivores, larvae are dependent on copepod prey (Shepherd and Packer 2006). Bluefish juveniles utilize estuary habitats, and temperature is a major driver of migration timing (Smith et al. 1994). When only these important factors were used in the exposure calculation, exposure was Moderate due to changes in zooplankton biomass exposure (Figure 10; Figure 11).

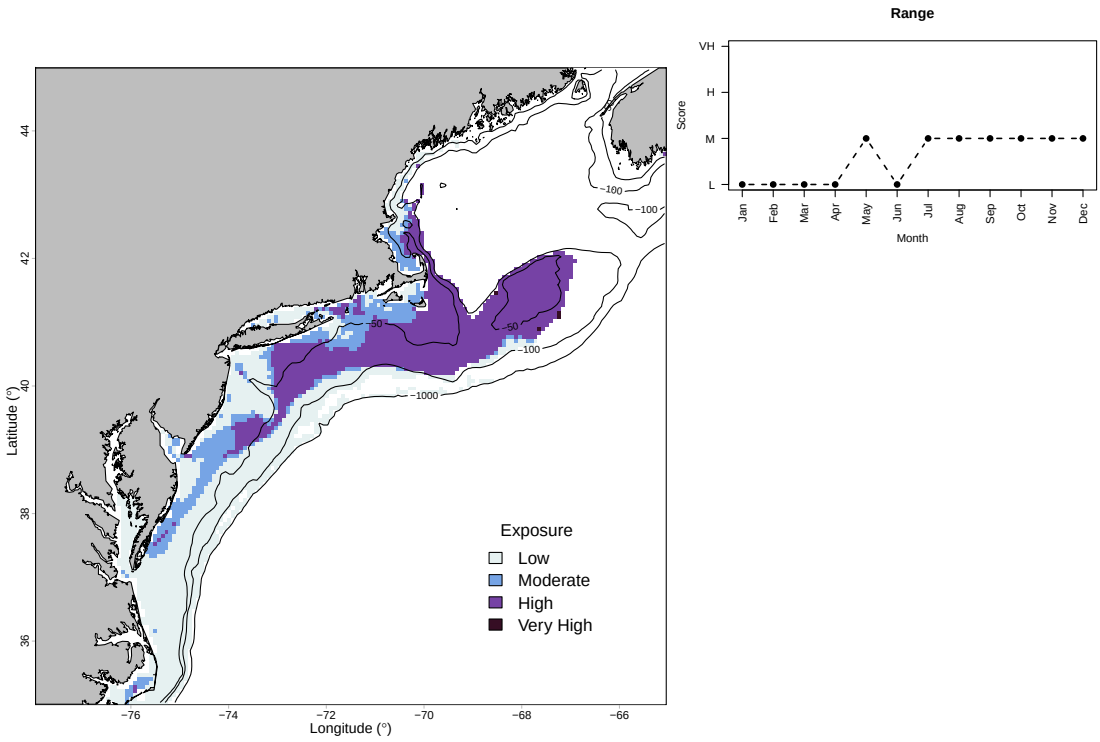


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Bluefish distributions

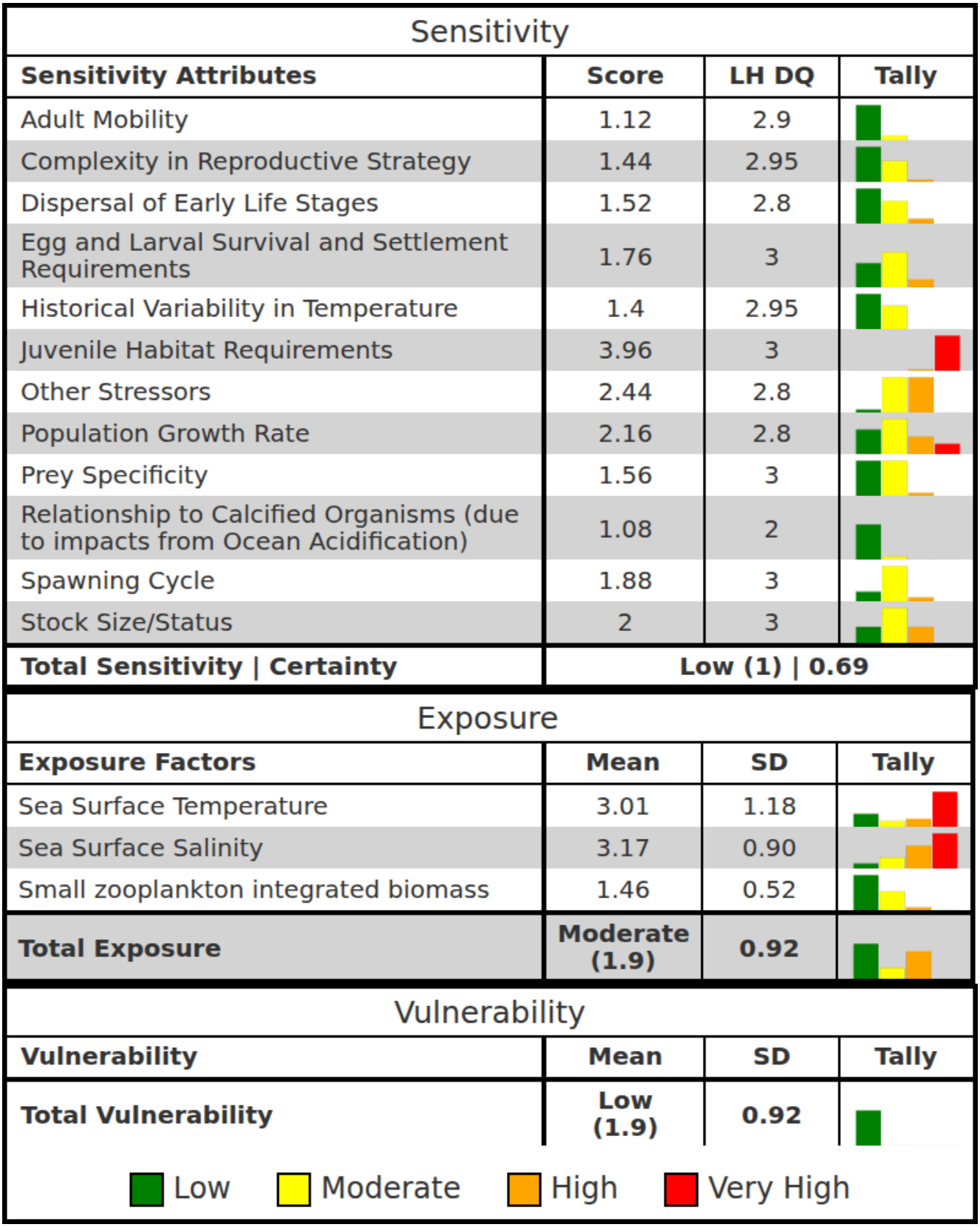


Figure 11: As in Figure 1 for Bluefish, but with only significant Exposure factors and the resulting Vulnerability Scores.

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